Some Important Testing Problems

(A) Y is just a white noise

$$H_0: \ \beta_0 = \beta_1 = \cdots = \beta_p = 0 \qquad A = I_{p+1}, \quad c = 0$$

$$RSS_{H_0} - RSS = \hat{\beta}_{\text{ols}}^T (X'X) \hat{\beta}_{\text{ols}} = \underset{\sim}{Y}^T X (X'X)^{-1} X^T \underset{\sim}{Y} = \underset{\sim}{Y}^T P(X) \underset{\sim}{Y} \sim \sigma^2 \chi^2_{p+1}$$

$$RSS = \underset{\sim}{Y}^T \big(I - P(X)\big) \underset{\sim}{Y} \sim \sigma^2 \chi^2_{n-p-1}$$

$$H_0: \ A\beta = c \qquad \frac{RSS_{H_0} - RSS}{r \cdot RSS} \overset{H_0}{\sim} F_{r, n-p-1}$$

$$RSS_{H_0} - RSS = \big(A\hat{\beta}_{\text{ols}} - c\big)^T \big[A(X'X)^{-1} A^T\big]^{-1} \big(A\hat{\beta}_{\text{ols}} - c\big)$$

independent of RSS.

(B) None of the covariates affect Y significantly

$$H_0: \ \beta_1 = \cdots = \beta_p = 0 \qquad A = \begin{bmatrix} 0 & : & I_p \end{bmatrix} \quad (p \times (p+1)), \qquad c = 0$$

Under H_0:

$$\underset{\sim}{Y} \overset{H_0}{\sim} N(\beta_0 \underset{\sim}{1}, \sigma^2 I)$$

$$\big[A(X^T X)^{-1} A^T\big]^{-1}$$

$$X \longrightarrow X_c = \begin{bmatrix} \underline{1} & X_{(1)} - \bar{x}_1 \underline{1} & \cdots & X_{(p)} - \bar{x}_p \underline{1} \end{bmatrix}$$

$$X_c^T X_c = \begin{bmatrix} n & 0^T \\ 0 & Z_c^T Z_c \end{bmatrix}$$

Log-likelihood:

$$\ell_n^*(\beta_0, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_0)^2$$

$$\frac{\partial \ell_n^*}{\partial \beta_0} = \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_0) = 0 \quad \Rightarrow \quad \hat{\beta}_0 = \bar{y}$$

$$\frac{\partial \ell_n^*}{\partial \sigma^2} = 0 \quad \Rightarrow \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2 \qquad (*)$$

$$\frac{\partial^2 \ell_n^*}{\partial \beta_0^2} = -n, \qquad \frac{\partial^2 \ell_n^*}{\partial \sigma^2\, \partial \beta_0} = -\frac{1}{\sigma^4}\sum_{i=1}^{n}(y_i - \beta_0) = 0 \quad \text{(at MLE)},$$

$$\frac{\partial^2 \ell_n^*}{\partial (\sigma^2)^2} < 0 \quad \text{(verify)}.$$

$$n\hat{\sigma}^2 \text{ in } (*) \text{ is } RSS_{H_0} = \sum_{i=1}^{n}(y_i - \bar{y})^2 = \underset{\sim}{Y}^T\left(I - \tfrac{1}{n}\underset{\sim}{1}\underset{\sim}{1}^T\right)\underset{\sim}{Y}$$

$$RSS = \underset{\sim}{Y}^T\big(I - P(X)\big)\underset{\sim}{Y}$$

$$RSS_{H_0} - RSS = \underset{\sim}{Y}^T\left(P(X) - \tfrac{1}{n}\underset{\sim}{1}\underset{\sim}{1}^T\right)\underset{\sim}{Y}$$

Recall

$$P(X) = P(X_c) = P\left(\underset{\sim}{1}\right) + P(Z_c)$$

where nxp

$$Z_c = \left[X_{(1)} - \bar{x}_1\,\underset{\sim}{1},\ \ldots,\ X_{(p)} - \bar{x}_p\,\underset{\sim}{1}\right]$$

So

$$RSS_{H_0} - RSS = \underset{\sim}{Y}^T P(Z_c)\underset{\sim}{Y} \sim \sigma^2\chi_p^2.$$

Coefficient of Determination

Multiple correlation coefficient (MCC): $Y, x_1, \ldots, x_p$.

$$r_{X_j,Y} \ :\ \text{Simple correlation coefficient}$$

$$R = \text{simple correlation between } Y \text{ and } \hat{Y} = P(X)\underset{\sim}{Y} = \frac{\sum_{i=1}^{n}(y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2\sum_{i=1}^{n}(\hat{y}_i - \bar{\hat{y}})^2}}$$

R^2 is called the **Coefficient of Determination**.

Correlation matrix

$$R^* = \big((r_{x_j,x_k})\big).$$

Multiple correlation coefficient R.

Partial correlation:

- Regress Y on $\{x_2, \ldots, x_p\}$; obtain residual $e_{Y\cdot 2\ldots p}$.
- Regress x_1 on $\{x_2, \ldots, x_p\}$; obtain residual $e_{x_1\cdot 2\ldots p}$.
- Finally, find the simple correlation between $e_{Y\cdot 2\ldots p}$ and $e_{x_1\cdot 2\ldots p}$.

$$Y_i = \beta_0 + \beta_1 x_{1i} + \varepsilon_i \beta_1 = r_{y,x} \qquad \text{(simple model)}$$

$$Y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_p x_{pi} + \varepsilon_i \qquad (\beta_1 = \text{partial correlation})$$

Result:

(a) $\bar{y} = \bar{\hat{y}}$

(b) $RSS_{H_0} = RSS + \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2$

(c) $R^2 = 1 - \dfrac{RSS}{RSS_{H_0}} = 1 - \big(\Lambda(\underset{\sim}{Y})\big)^{2/n}$

where $\Lambda(\underset{\sim}{Y})$ is the LRT test statistic.

Proof:

(a)

$$\bar{y} = \tfrac{1}{n}\underset{\sim}{1}^T\underset{\sim}{Y}, \qquad \bar{\hat{y}} = \tfrac{1}{n}\underset{\sim}{1}^T\hat{\underset{\sim}{Y}} = \tfrac{1}{n}\underset{\sim}{1}^T P(X)\underset{\sim}{Y} = \tfrac{1}{n}\underset{\sim}{1}^T\underset{\sim}{Y} = \bar{y}$$

because

$$P(X) = P(X_c) = \tfrac{1}{n}\underset{\sim}{1}\underset{\sim}{1}^T + P(Z_c) \quad \text{and} \quad \underset{\sim}{1}^T P(X) = \underset{\sim}{1}^T, \quad \underset{\sim}{1}^T Z_c = 0.$$

(b)

$$\begin{aligned}
\text{RHS} = RSS + \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 &= \underset{\sim}{Y}^T\big(I - P(X)\big)\underset{\sim}{Y} + \underset{\sim}{Y}^T P(X)\big(I - \tfrac{1}{n}\underset{\sim}{1}\underset{\sim}{1}^T\big)P(X)\underset{\sim}{Y} \\
&= \underset{\sim}{Y}^T\Big(I - P(X) + P(X) - P(X)\tfrac{1}{n}\underset{\sim}{1}\underset{\sim}{1}^T P(X)\Big)\underset{\sim}{Y} \\
&= \underset{\sim}{Y}^T\Big(I - \tfrac{1}{n}\underset{\sim}{1}\underset{\sim}{1}^T\Big)\underset{\sim}{Y} = \sum_{i=1}^{n}(y_i - \bar{y})^2.
\end{aligned}$$

(Note: $(\underset{\sim}{1}) \subseteq \mathcal{C}(X)$.)

$$X = \big[X_{(1)} : X_{(2)}\big]$$

$$P(X)\,P(X_{(1)}) = P(X_{(1)})$$

$$\text{As} \qquad \underbrace{P(x)X_{(1)}}_{X_{(1)}}\Big(X_{(1)}^\top X_{(1)}\Big)^{-1}X_{(1)}^\top = P\big(X_{(1)}\big)$$

(c) Observe that

$$\begin{aligned}
\sum_{i=1}^{n}(y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}}) &= \big\langle \big(I - \tfrac{1}{n}\underset{\sim}{1}\underset{\sim}{1}^T\big)\underset{\sim}{Y},\ \big(I - \tfrac{1}{n}\underset{\sim}{1}\underset{\sim}{1}^T\big)P(X)\underset{\sim}{Y}\big\rangle \\
&= \underset{\sim}{Y}^T\big(I - \tfrac{1}{n}\underset{\sim}{1}\underset{\sim}{1}^T\big)\big(P(X) - \tfrac{1}{n}\underset{\sim}{1}\underset{\sim}{1}^T\big)\underset{\sim}{Y} \\
&= \underset{\sim}{Y}^T\big(P(X) - \tfrac{1}{n}\underset{\sim}{1}\underset{\sim}{1}^T\big)\underset{\sim}{Y} = \big\|\big(P(X) - P(\underset{\sim}{1})\big)\underset{\sim}{Y}\big\|^2 = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2.
\end{aligned}$$

Thus

$$R^2 = \frac{\big[\sum(\hat{y}_i - \bar{y})^2\big]^2}{\big[\sum(\hat{y}_i - \bar{y})^2\big]\big[\sum(y_i - \bar{y})^2\big]} = \frac{\sum(\hat{y}_i - \bar{y})^2}{\sum(y_i - \bar{y})^2} = 1 - \frac{RSS}{RSS_{H_0}}.$$

(C) Some of the variables are not important

$$X_n = \left[X_{(1)} : X_{(2)}\right] = [\underbrace{\underset{\sim}{1},\ X_{(1)}, \ldots, X_{(q)}}_{X_{(1)}}, \underbrace{X_{(q+1)}, \ldots, X_{(p)}}_{X_{(2)}}]$$

$$H_0 :\ \beta_{q+1} = \cdots = \beta_p = 0 \quad \Leftrightarrow \quad \beta_{q+1} = 0,\ \beta_{q+2} = 0,\ \ldots,\ \beta_p = 0$$

$$H_1 :\ \text{No restriction}$$

$$A = \begin{bmatrix} 0 & I_{p-q} \end{bmatrix} \quad \big((p-q) \times (q+1)\big), \qquad c = 0$$

Degrees of freedom: $p + 1 - q - 1 = p - q$.

$$\text{Adjusted } R^2 = \frac{R^2}{\#\text{ of regressors}}.$$

Calculate $RSS_{H_0} - RSS$ using formula (Option 2):

$$\ell_{H_0}\big((\beta_0, \beta_1, \ldots, \beta_q)', \sigma^2\big) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\left\|\underset{\sim}{Y} - X_{(1)}\beta_{(1)}\right\|^2$$

$$RSS_{H_0} = \underset{\sim}{Y}^T\big(I - P(X_{(1)})\big)\underset{\sim}{Y}, \qquad RSS = \underset{\sim}{Y}^T\big(I - P(X_n)\big)\underset{\sim}{Y}$$

$$\frac{RSS_{H_0} - RSS}{RSS} = \frac{\underset{\sim}{Y}^T\big(P(X_n) - P(X_{(1)})\big)\underset{\sim}{Y}}{\underset{\sim}{Y}^T\big(I - P(X_n)\big)\underset{\sim}{Y}} \overset{H_0}{\sim} F_{p-q,\, n-p-1}$$

(verify).

$$\left[P(X_n) - P(X_{(1)})\right]^2 = P^2(X_n) + P^2(X_{(1)}) - P(X_n)P(X_{(1)}) - P(X_{(1)})P(X_n) = P(X_n) - P(X_{(1)})$$

(as $\mathcal{C}(X_{(1)}) \subseteq \mathcal{C}(X)$).

$$P(X_{(1)})P(X) = X_{(1)}(X_{(1)}^T X_{(1)})^{-1}X_{(1)}^T P(X) = P(X_{(1)})$$

because $P(X)X_{(1)} = X_{(1)}$.

(D) Effects of all the covariates are same

$$H_0 :\ \beta_1 = \beta_2 = \cdots = \beta_p \qquad H_1 :\ \text{No restriction}$$

$$\beta_1 - \beta_2 = 0,$$
$$\vdots$$
$$\beta_{p-1} - \beta_p = 0.$$

$$A = \begin{bmatrix} 0 & 1 & -1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & -1 & \cdots & 0 \\ \vdots & & & \ddots & \ddots & \\ 0 & 0 & \cdots & 0 & 1 & -1 \end{bmatrix} \quad \big((p-1)\times(p+1)\big), \qquad c = 0.$$

$$\ell_{H_0}(\beta_0, \beta, \sigma^2) = -\frac{n}{2}\log\sigma^2 - \frac{n}{2}\log(2\pi) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}\big(y_i - \beta_0 - \beta(x_{1i} + \cdots + x_{pi})\big)^2$$

Homework: Complete this.

$$Z = X_{(1)} + \cdots + X_{(p)}, \qquad X_{H_0} = \big[\underset{\sim}{1},\, Z\big]$$

$$RSS_{H_0} = \underset{\sim}{Y}^T\big(I - P(X_{H_0})\big)\underset{\sim}{Y}, \qquad \mathcal{C}(X_{H_0}) \subseteq \mathcal{C}(X_n).$$

(E) Sum of regression coefficients is k

$$H_0:\ \beta_1 + \cdots + \beta_p = k$$

$$A \text{ is a vector, say } \underset{\sim}{a} = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 1 \end{pmatrix}, \qquad c = k.$$

$$RSS_{H_0} - RSS = \frac{(\underset{\sim}{a}^T\hat{\beta} - k)^2}{\underset{\sim}{a}^T(X'X)^{-1}\underset{\sim}{a}}$$

$$(X'X)^{-1} = \begin{bmatrix} v_{00} & v_{01}^T \\ v_{01} & V_{(2)} \end{bmatrix}, \qquad \underset{\sim}{a}^T(X'X)^{-1}\underset{\sim}{a} = \underset{\sim}{1}^T V_{(2)}\underset{\sim}{1}.$$

$$\frac{RSS_{H_0} - RSS}{RSS} \sim F_{1,n-p-1} \equiv t^2_{n-p-1}.$$

Interval Estimation

Univariate:

$$\big[L(\underset{\sim}{Y}),\, U(\underset{\sim}{Y})\big] \quad \text{pair of statistics.}$$

$$P_\beta\big(L(\underset{\sim}{Y}) \le a^T\beta \le U(\underset{\sim}{Y})\big) \ge (1-\alpha).$$

Multivariate:

$$S(\underset{\sim}{Y}): \text{ Random set which is a function of } \underset{\sim}{Y}.$$

$$P_\beta\big(\beta \in S(\underset{\sim}{Y})\big) \geq (1-\alpha).$$

More generally, $g : \mathbb{R}^p \to \mathbb{R}^k$,

$$P_\beta\big(g(\beta) \in S(\underset{\sim}{Y})\big) \geq (1-\alpha).$$

Example:

$$g(\beta) = \begin{pmatrix} \beta_1 - \beta_2 \\ \vdots \\ \beta_p - \beta_{p-1} \end{pmatrix} = A\beta.$$

Confidence Ellipsoid

Let A be an $r \times (p+1)$ matrix of rank r.

$$A\hat{\beta}_{\text{ols}} \;\sim\; N_r\big(A\beta,\; \sigma^2 A(X'X)^{-1}A^T(\textit{invertible})\big)$$

$$\frac{1}{\sigma}\big[A(X'X)^{-1}A^T\big]^{-1/2}(A\hat{\beta}_{\text{ols}} - A\beta) \sim N_r(0, I)$$

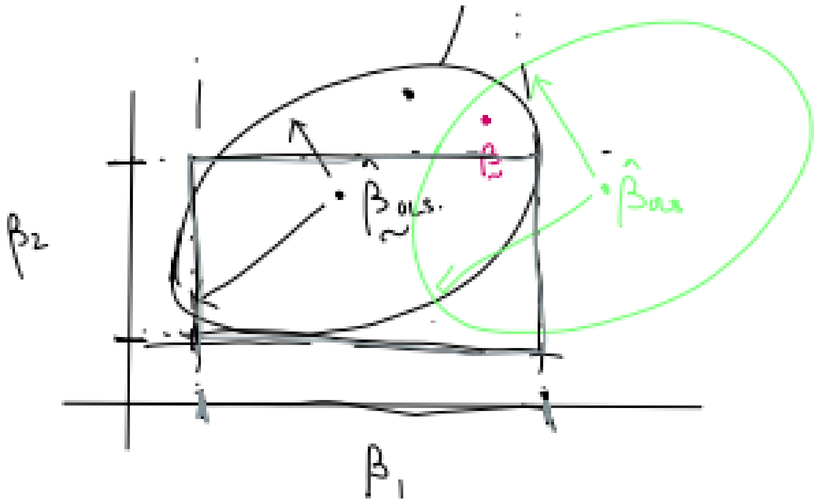
$$\frac{1}{\sigma^2}(A\hat{\beta}_{\text{ols}} - A\beta)^T\big[A(X'X)^{-1}A^T\big]^{-1}(A\hat{\beta}_{\text{ols}} - A\beta) \sim \chi^2_r$$

$$\frac{RSS}{\sigma^2} \sim \chi^2_{n-p-1} \quad \text{independent of } \hat{\beta}_{\text{ols}} \perp\!\!\!\perp RSS.$$

$$\frac{\frac{1}{r}(A\hat{\beta}_{\text{ols}} - A\beta)^T\big[A(X'X)^{-1}A^T\big]^{-1}(A\hat{\beta}_{\text{ols}} - A\beta)}{RSS/(n-p-1)} \sim F_{r,\,n-p-1}.$$

$$P_\beta\left(\frac{n-p-1}{r \cdot RSS}(\hat{\beta}_{\text{ols}} - \beta)^T A^T\big[A(X'X)^{-1}A^T\big]^{-1}A(\hat{\beta}_{\text{ols}} - \beta) \leq F_{\alpha;\,r,\,n-p-1}\right) = (1-\alpha).$$

(The quadratic form is denoted Ψ.)



What if

$$A = \begin{bmatrix} a_1^T \\ \vdots \\ a_r^T \end{bmatrix}?$$

Suppose we find individual CIs for $a_j^T\beta$, say $[L_j(\underset{\sim}{Y}), U_j(\underset{\sim}{Y})]$, $j = 1, \ldots, r$, and consider the intersection

$$P_\beta\big(L_1 \le a_1^T\beta \le U_1,\ \ldots,\ L_r \le a_r^T\beta \le U_r\big).$$

$$P_\beta\big(L_1 \le a_1^T\beta \le U_1\big) = (1-\alpha).$$